

Support Systems and Well-Being: How Connection Affects Life Satisfaction and Loneliness in Difficult Times (COVID-19)

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Goal

Since COVID-19 significantly disrupted social interactions and increased stress, loneliness, and mental health challenges (Canada, 2024).

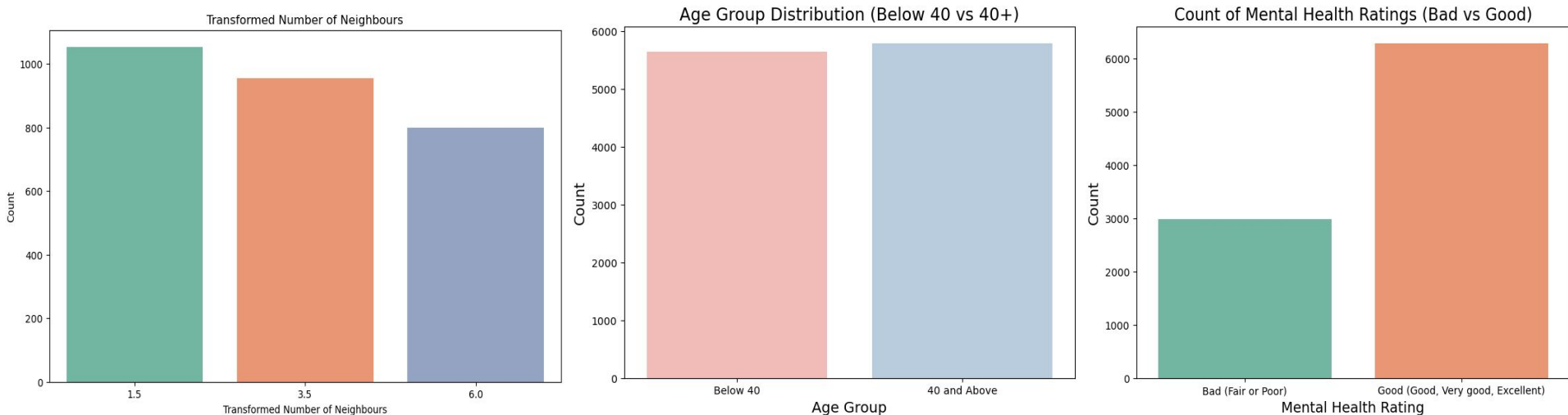
See how quality of life through loneliness ratings and life satisfaction ratings can be improved based on social connections and interactions.

Use data from Canadian Social Connections Survey to explore how community connections (knowing neighbours by name, owning dogs, living with family) help raise life satisfaction ratings during times of crises (COVID-19)

"Is there a relationship between the number of neighbours someone knows by name and their self-rated mental well-being during the pandemic, and does this relationship differ across age groups?"

Why this Question?

- The role of social ties in promoting mental well-being (Kawachi, 2001) is important.
- Knowing neighbors by name is a practical measure of localized social connection (Ohta et al., 2024).
- Social needs and resources often vary across age groups - guide age-specific interventions.
 - Younger individuals may rely more on peer networks outside the neighborhood (Orben, Tomova and Blakemore, 2020),
 - Older adults need more proximal connections due to mobility or lifestyle changes (Gardner, 2008)
- Could potentially inform public health strategies (Sandel et al., 2016).



Variables	Predictor: Number of neighbours someone knows by name	Predictor: Age of respondents	Interaction: between Number of neighbours known and Age	Outcome: Self-rated mental health of respondents
Classification → Treatment	Ordinal categorical → Discrete numerical variable	Continuous → Binary indicator	Multiplying the predictor variables	Ordinal categorical → Binary categorical variable

Set up of Analysis Methodology examine relationship between predictor variables and the binary outcome variable `mental_health_binary` (1 = "good" and 0 = "bad")

1. **Data Preprocessing:**

- **Selection of Relevant Columns**
- **Data Cleaning:** Dropped "Presented but no response", missing values (`NaN`) handled by replacing them appropriately

2. **Variable Transformation:**

- **Categorical to Numerical Mapping:** Number of Neighbours Known
- **Mental Health Mapping:** converted into a binary variable
 - simplify analysis and facilitate logistic regression modeling
- **Age Binary Variable:** A new binary variable `age_binary` created (Britannica, 1998).

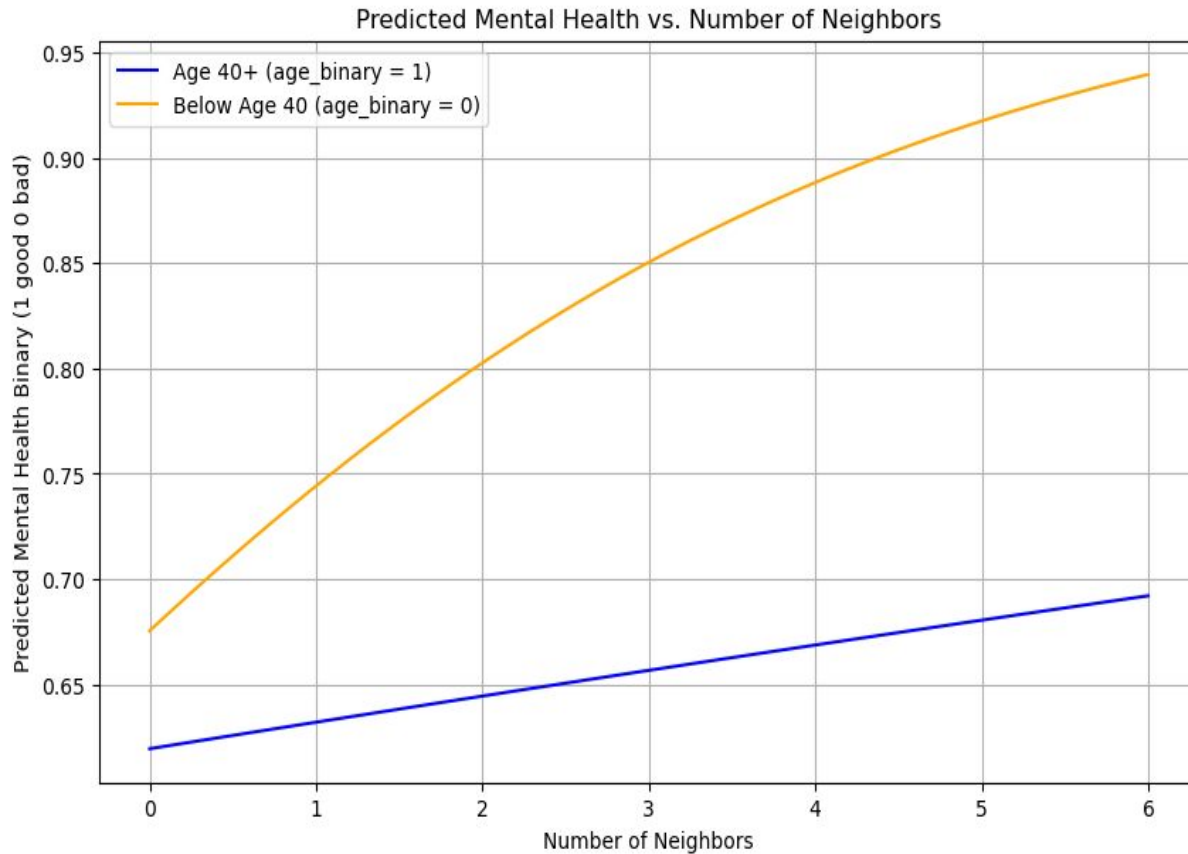
3. **Interaction Term:**

- between the Number of Neighbours Known and Age Binary variables is created to explore if the relationship between the number of neighbors and mental health differs by age group

4. **Logistic Regression Model:**

- A logistic regression model is fitted using the formula
- This model predicts the binary outcome (`mental_health_binary`) based on the predictors

Results



Coefficients:

- **Intercept:** (0.7330) significantly different from zero, indicating a baseline level is not 0
- **Number of neighbours:** (0.3349) is positive suggesting that a higher number of neighbors is associated with better mental health
- **Age Binary:** (-0.2454) is negative and significant indicating that being 40+ is associated with a lower likelihood of good mental health
- **Interaction:** (-0.2811) is negative suggesting that the effect of number of neighbours on mental health is moderated by age, the effect being less pronounced for individuals 40+.

- **Coefficients and Significance:** All predictors, including the interaction term, are statistically significant (p values < 0.01). unlikely due to chance

Interpretation

1. **Social Connections:** The number of neighbors individuals can name has a **positive and significant impact on mental health** for all ages, effect is **stronger for younger individuals**.
2. **Age and Social Connection Interaction:** Older adults **benefit less** from social connections in terms of improving mental health. Reflects **differing social needs** or barriers to utilizing social connections as a mental health resource.

Further exploration: The model explains a modest proportion of the variance in mental health (3.25%), and the significant predictors suggest important factors that could be further explored in future analyses.

Implications

- **Public Health Interventions:** Mental health enhancement efforts might focus on fostering social connections, **particularly among younger individuals**, we should considering unique strategies to support older adults.
- **Community Design:** Creating opportunities for older adults to engage in meaningful relationships could help mitigate the weaker relationship between neighbors and mental health in this group.

Did dogs help people with loneliness during the changes because of Covid? How do they compare to people without dogs?

Why do we ask this question?

- Observing loneliness during COVID and owning a dog(s) during COVID can show the impact of dogs on loneliness and raise awareness about well-being

Why this matters:

- Dogs are companions and can mitigate loneliness. By focusing on loneliness level changes for dog owners vs. non dog owners, we can provide evidence for this.

Background Information (a study on Canadian pet owners during COVID)

- Pet owners have lower loneliness scores compared to non-pet owners, supporting the correlation between pet ownership and improved mental health
- Pet ownership is especially beneficial for people who live alone

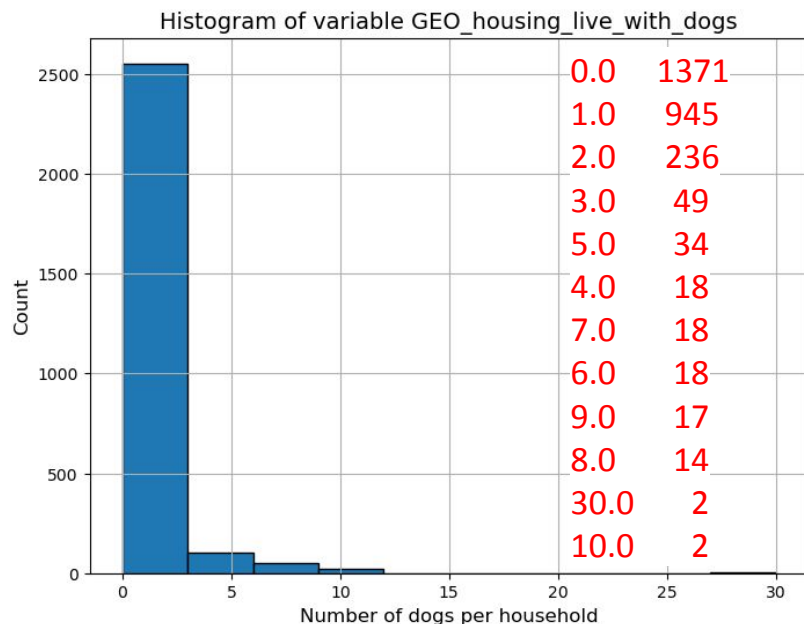


Figure 1

Numerical variables:

➤ GEO_housing_live_with_dogs

What does it do?

➤ Counts the number of dogs per household

*Refer to Figure 1

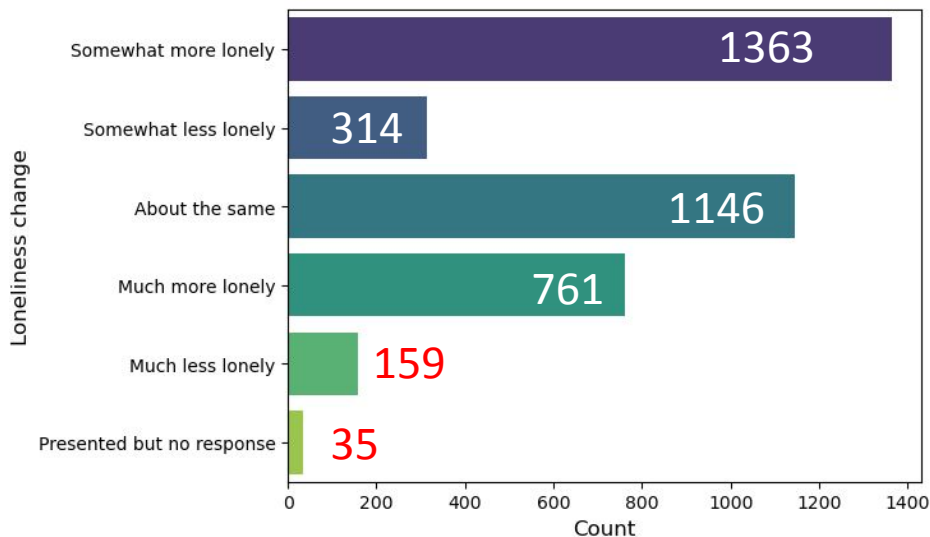


Figure 2

Categorical variables:

➤ LONELY_change_pre_covid

What does it do?

➤ Compares participants' loneliness right now to before COVID

*Refer to Figure 2

Visualization Methodology

Data cleaning

- **Remove null values:** to ensure the dataset was complete and reliable for analysis.
- **Filter out outliers:** Removed households with more than 5 dogs (`GEO_housing_live_with_dogs`) as these were considered anomalies.

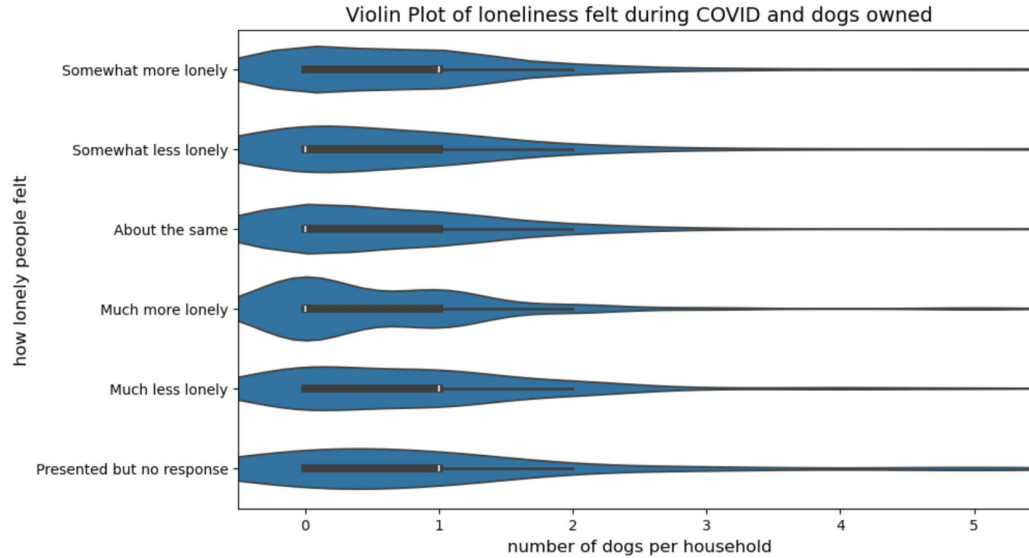
Use cleaned data:

- Analyzed the relationship between the number of dogs per household (`GEO_housing_live_with_dogs`) and loneliness change scores during COVID-19 (`LONELY_change_pre_covid`).
- Explored these variables through summary statistics and visualizations to assess potential trends.

Create visualizations:

- Generate a violin plot to visualize how changes in loneliness during COVID-19 varied by the number of dogs in households.
- A simple plot that shows the distribution and is interpreted by its shape (wideness)
- Focus on **much more lonely** category to draw conclusions

Removed 70 outliers where the number of dogs exceeded 5 per household. These are considered anomalies.



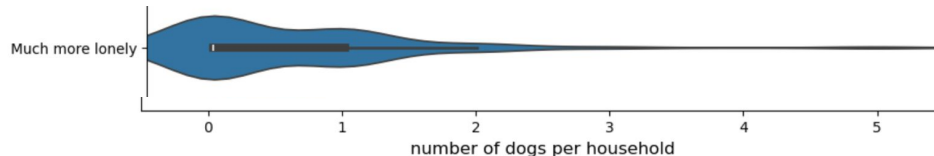
*Median indicated by white bar

* Plots are positive skewed
meaning average > median

- Look at individual plots; compare density (the width) for **much more lonely** plot
 - Wider the plot, the more responses are concentrated
 - Most dense around 0 number of dogs (category: much more lonely)
 - Density suggests that many people (relative to other values) reported those scores → people with zero dogs reported feeling more lonely

Findings and implications

- Based on the **much more lonely** violin plot, people with zero dogs reported feeling **more lonely**
- The violin is wide at 0 dogs per household
 - Meaning more people in this group felt this level of loneliness (much more lonely)
- It gets narrow at 1 dog per household
 - Meaning fewer people felt much more lonely with at least 1 dog in the house
- Suggesting that owning dogs **CAN** alleviate loneliness scores during times of isolation
- **Next steps**
 - Focus on more dog oriented therapeutic practices or coping strategies to help those who feel lonely
 - I.e. therapy dogs, emotional support animals

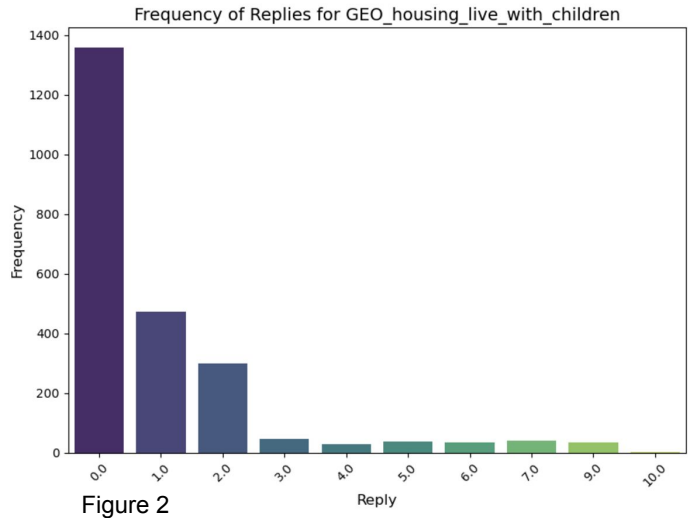
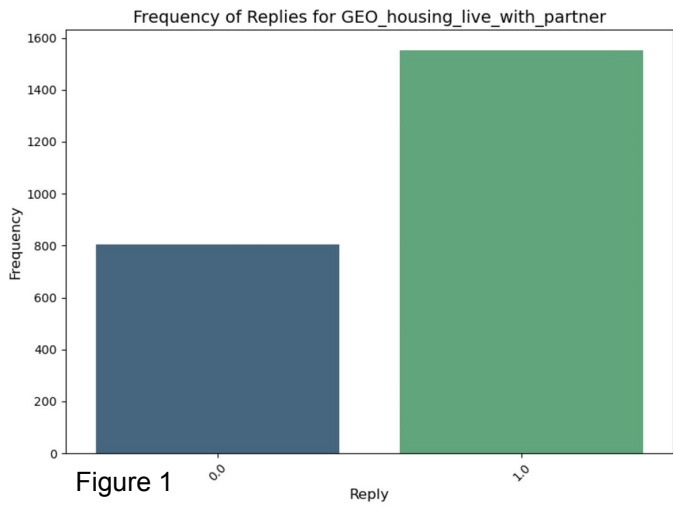


What is the overall effect on general well-being of living with a partner/kids during the pandemic? What is the effect on specific aspects of wellbeing(i.e depression, loneliness, exhaustion)?

Why this Question?

- Restrictions being put on people during the pandemic made it difficult to leave the house and dictated who you spent time with
- Loneliness and depression often had to be combatted from inside the house
- More big decisions surrounding relationships were made during the pandemic with 3 in 10 Canadians saying the pandemic made them move in with their significant other (Thompson, 2020)

It is important to know how relationships within the house affect people so we looked at the relationship between life satisfaction and living with kids/partners



Predictor Variables:

- Living with **children**, binary data, and living with a **partner**, binary data
- Many people didn't respond to this section of the survey

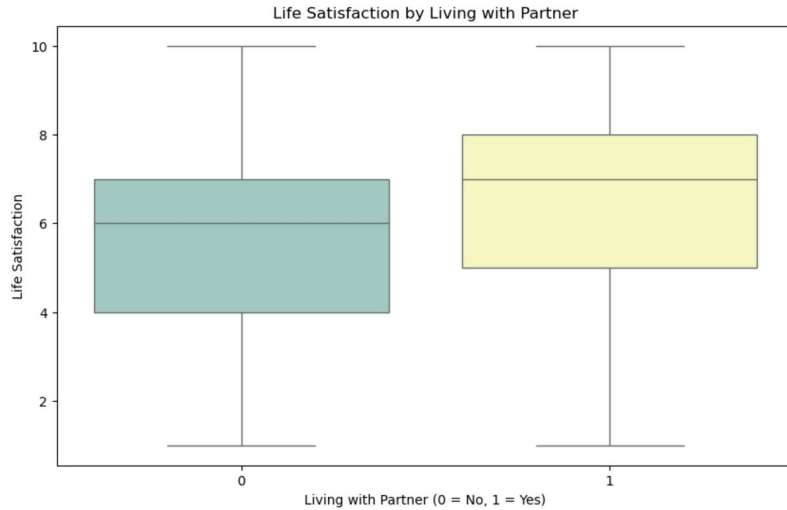
Dependent Variable:

- **life satisfaction**, discrete data on a 1-10 scale
- **10 separate wellness variables**, categorical rankings based on feelings of hopelessness, sickness, and other feelings

Analysis and Methodology

1. Remove null values
2. Turn children and partner into binary data
3. Use this data as predictors in multiple linear regression to find relationship
4. Make visualizations to show that the data reflects these relationships

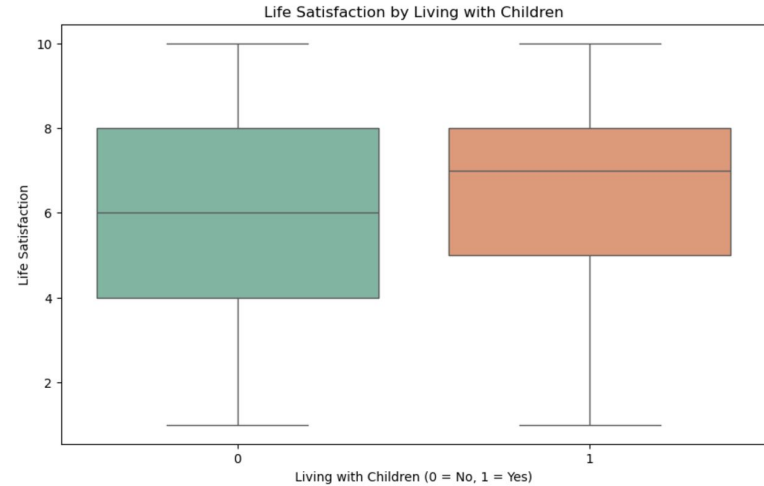
Figure 1



From regression model, living with a partner during the pandemic increased the individuals life satisfaction score by 0.6 on average.

In Figure 1 you can see that in general people were more satisfied with life if they had a partner they were living with.

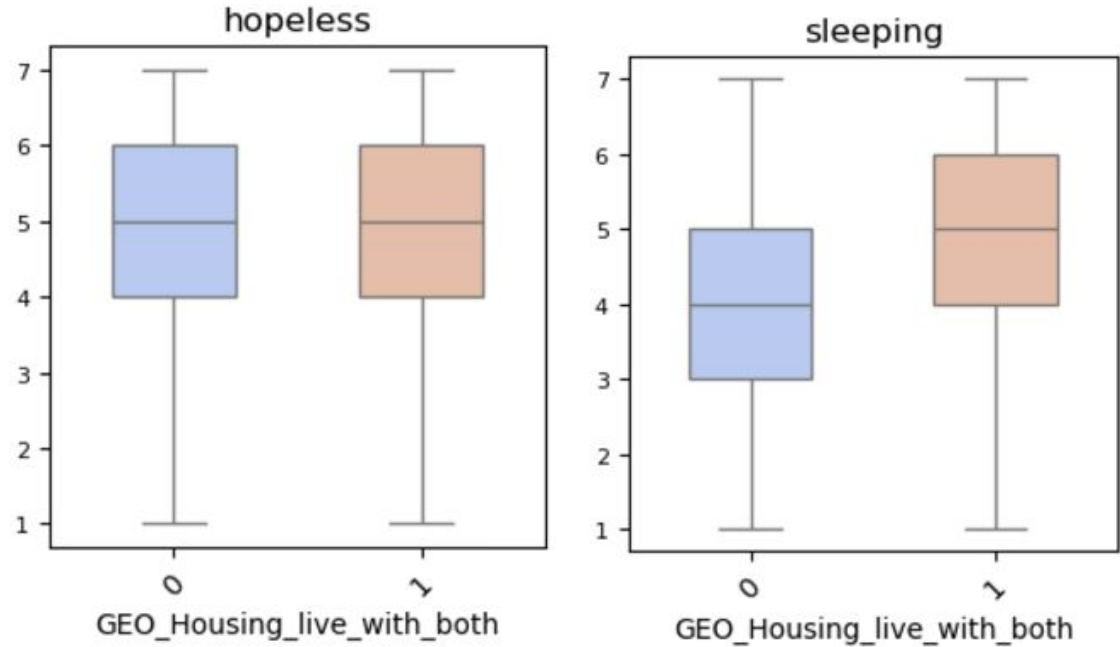
Figure 2



Compared to figure 1 the difference in figure 2 is less noticeable.

Children still had a positive affect, with a 0.3 ranking increase, but it has less statistical evidence to back it up and the range of life satisfaction ranking is very close to someone living without kids.

- Separate elements of life satisfaction were almost exclusively made better by having a partner(average increase between 0.2 and 0.6)
- Effect of children did not have as much evidence and was not strictly positive(most p values >0.05)
- Overall having a partner seemed to have the most positive effects



Y axis scale is 1-7 with 1 feeling this emotion always and 7 not at all. This might seem backwards but it is to keep numerical positive changes as positive changes in real life

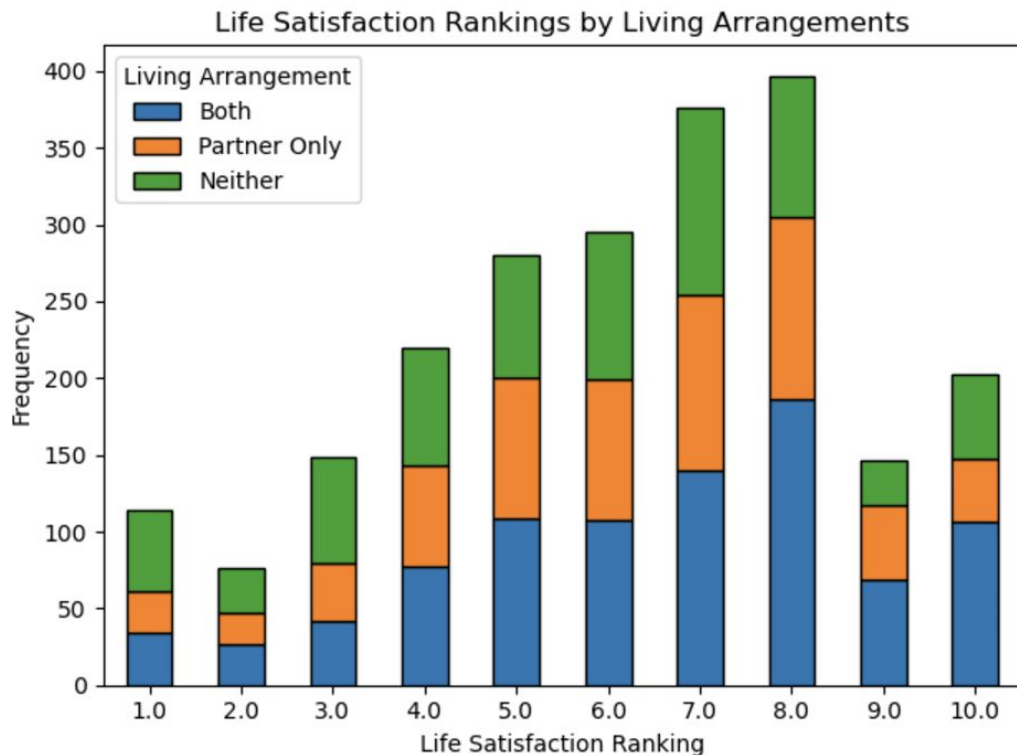
Interpretation and Implications

Living with a partner had a **positive effect** on an individual's life satisfaction.

Living with children had a **positive effect** on an individual's life satisfaction but less of an effect compared to partners, children come with increased stress (Gadermann et al., 2021)

Help and **give support** to people with less/no family

Focus on **building healthy familial relationships** so that individuals can find this increase in life satisfaction



Limitations: how our models and processes can be improved

Binary Transformation: Converting nuanced data into binary categories can oversimplify and obscure subtle trends or patterns in the data.

Age Grouping and Middle-Aged Reference: The choice of 40 as the midpoint of the life course may not account for diverse mental health and social support needs across different age groups, leading to potential misrepresentation.

Arbitrary Grouping of Connection Numbers: Grouping variables, such as the number of neighbors known by name, into arbitrary ranges may limit the model's ability to detect meaningful variations or relationships.

Subjectivity in Self-Rated Wellness: Self-reported wellness measures are subject to interpretation, which can introduce inconsistencies and variability in the data.

Model Specification: The model's simplicity may fail to capture complex relationships or interactions between variables.

Survey Bias: Self-reported data on mental health and social behaviors are prone to recall bias, social desirability bias, or other inaccuracies.

Sample Bias: If the sample is not representative of the general population, the findings may lack generalizability.

Missing Data: Partial responses (e.g., reporting household dog ownership but not loneliness changes) can skew medians, averages, or category-specific insights.

Conclusion

Overall result from 3 research questions:

- Neighbors, family, and pets are valid support systems for reducing loneliness and life satisfaction during COVID
- We provided evidence for this by using CSCS data and statistical knowledge
- Further analysis suggestions: Analyzing the relationship between well-being and other alleged support systems
- Advice for the public: To reduce isolation, you can reach out to support groups

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